

Knowledge-Guided Generative Re-Stylization of CIM Models with Feature Continuance

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Abstract—As a core enabler of digital twin cities, City Information Modeling (CIM) faces dual challenges in multi-party data sharing and application: ensuring privacy and security while fulfilling the growing demand for diversified digital assets. On one hand, high-precision 3D city models expose sensitive personal and public information. On the other, singular data sources struggle to satisfy the demand for diverse digital assets in fields like virtual simulation and content creation. Existing data desensitization methods, such as blurring or simple procedural replacement, universally sacrifice visual realism and semantic integrity, severely diminishing the core value of CIM data. To address this dilemma, this research introduces a novel paradigm: "Knowledge-Guided Feature-Continuous Regeneration." The core idea is to leverage generative AI to completely reshape a building's geometry and appearance, thereby severing the direct link to original sensitive data. Simultaneously, a structured "visual knowledge fingerprint" vector enables the controllable inheritance of the original building's core features, such as physical constraints, semantic composition, and macro-style. This paper introduces a four-stage hybrid AI generation pipeline, 'Deconstruct-Generate-Reconstruct-Render', leveraging a purpose-built architectural layout generation model (Arch-Layout LoRA) and a knowledge-driven procedural engine. Experiments demonstrate that this method exhibits significant advantages in controllability, realism, and diversity. It not only provides a "usable-but-unseen" security solution for CIM data sharing but also pioneers a new path for transforming single data sources into diverse digital assets.

Keywords—City Information Modeling (CIM), Data Privacy, Diffusion Models, Digital Asset Generation, Generative AI, Procedural Modeling.

I. INTRODUCTION

WITH THE ACCELERATION of urbanization and the rapid advancement of information technology, digital twins and smart cities have evolved from forward-looking concepts into core technologies supporting urban planning, management, and operations. As the digital bedrock of this grand vision, City Information Modeling (CIM) integrates the strengths of Geographic Information Systems (GIS) and Building Information Modeling (BIM) [1]. It uses fine-grained 3D models to represent a city's physical form, semantic functions, and even dynamic processes, providing unprecedented data fidelity for critical applications such as traffic simulation, emergency management, and energy analysis.

A. The Digital Twin Dilemma: Balancing Utility and Security

A profound gap exists between the immense value of CIM data and its widespread application, stemming from the severe challenges of data sharing. On one hand, openness and collaboration are essential for unlocking the potential of data and fostering innovative applications. On the other hand, high-fidelity CIM models are themselves a collection of sensitive information. They may contain visual information that directly infringes on personal privacy (e.g., interior layouts visible through windows) and can also expose the layout of facilities critical to public safety (e.g., government buildings, power stations). This inherent contradiction, where the data's value and its risks both originate from its granularity, constitutes the

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core dilemma of the digital twin era. How to promote the circulation and reuse of CIM data while ensuring data security and privacy has become a key bottleneck constraining the development of smart cities.

B. Limitations of Existing "Subtractive" Desensitization Methods

To address the aforementioned challenges, academia and industry have explored various data protection technologies. However, these methods, when applied to complex 3D city models, can be fundamentally categorized as "subtractive" strategies that reduce risk by removing, obscuring, or simplifying information, thereby creating a direct trade-off between security and utility. The most direct approach, visual obfuscation, employs techniques such as blurring or pixelating building facades. This approach is inherently destructive, severely compromising the model's visual realism and aesthetic value, rendering the desensitized data useless for the vast majority of applications that rely on visual presentation, such as virtual city tours or film production. More semantically aware methods like data masking and suppression, widely used in the database domain, remove specific building components or attributes. Their core flaw is the direct loss of data utility; a building model lacking window details cannot be used for solar analysis, and an urban district model with generalized attributes cannot support fine-grained functional planning. Furthermore, such anonymization does not guarantee absolute security, as sensitive information can still be re-identified through linkage attacks. As a more advanced strategy, simple procedural replacement uses procedural generation to replace original buildings with structurally similar virtual ones. While this severs the direct geometric link to the original model, traditional rule-based procedural methods suffer from a creativity bottleneck. The generated buildings are often stylistically monotonous, with rigid details, and lack the rich diversity and subtle variations of real-world architecture, failing to meet the demand for high-quality visual content. As summarized in Table I, existing approaches force a compromise where security is achieved at the cost of data value.

TABLE I. COMPARATIVE ANALYSIS OF DATA ANONYMIZATION TECHNIQUES

<i>Technique</i>	<i>Visual Fidelity</i>	<i>Generative Diversity</i>	<i>Data Utility</i>	<i>Controllability</i>	<i>Feature Continuanace</i>
<i>Visual Obfuscation</i>	Extremely Low	None	Extremely Low	None	None
<i>Simple Procedural Replacement</i>	Low-Medium	Low	Medium	Low	Low
<i>Proposed Method (Ours)</i>	High	High	High	High	High

C. Our Contribution: A Generative Re-creation Framework

To transcend the limitations of existing methods, this paper proposes a paradigm shift: moving from "subtractive" information destruction to "generative" value re-creation. This work formally introduces a hybrid generation pipeline based on

four stages — "Deconstruct-Generate-Reconstruct-Render" — aimed at achieving knowledge-guided, feature-continuous regeneration of CIM models. The core of this method lies not in modifying the original data but in abstracting it into a 'visual knowledge fingerprint', using this 'genetic' blueprint to generate a new digital asset that is geometrically and visually distinct, yet architecturally continuous with the original. This framework transforms data security from a passive defense mechanism into an active, value-creating process of digital asset evolution. The core contributions are as follows: a complete framework for extracting, encoding, and inheriting an architectural 'visual knowledge fingerprint' as the cornerstone for feature continuity; a dedicated AI toolchain for architectural layout generation, including a custom LoRA model and evaluation metrics; a novel hybrid paradigm that combines AI's creative guidance with the precision of procedural modeling; and a multi-modal, knowledge-guided texture generation mechanism that fuses structural, semantic, and realistic information for high-fidelity appearance creation.

II. RELATED WORK

This research is lies at the intersection of several fields, including 3D urban data processing, 2D layout generation, 3D content generation, and texture stylization. This section reviews the state-of-the-art in each of each area to clarify the technical positioning and novelty of our work.

A. 3D Urban Data Privacy and Desensitization

Early privacy protection for 3D city models relied on geometric simplification and obfuscation. Procedural modeling was considered a superior desensitization method for its ability to generate entirely new geometries [4,5]. However, traditional procedural methods are "rule-driven," relying on expert-defined rules that struggle to capture the infinite variations of real-world architectural styles, resulting in insufficient diversity and realism. In contrast, our proposed method is "data-driven," leveraging deep learning models to learn intrinsic patterns from large-scale real architectural data to generate more natural and diverse results.

B. 2D Layout Generation

In 2D layout generation, models like GANs have achieved success in image translation but suffer from training instability and weak structural control [6,7]. While Graph Neural Networks (GNNs) better handle topological constraints for interior layout design, their strong constraints can limit form diversity for external building footprint generation [8,9], where preserving macro-form style is more critical. Recently, diffusion models with ControlNet have achieved breakthrough progress [10], enabling precise spatial guidance without compromising the capabilities of pre-trained large models, which aligns perfectly with our project's needs.

C. 3D Content Generation

End-to-end methods that directly generate 3D models from text or images, such as Shap-E [11], still struggle with architectural models requiring complex topology and precise planar structures, often producing low-polygon, ambiguous "blobby" shapes. Another class of "2D-Lifting" methods (e.g., DreamFusion [12, 13]), while capable of producing visually stunning objects, often disregards architectural priors like

planarity and orthogonality and faces challenges like slow convergence and the "Janus problem" (inconsistent multi-faceted views). Given the limitations of current end-to-end 3D generation, we chose a "procedural reconstruction" approach, which offers unparalleled advantages in ensuring geometric accuracy, topological correctness, and adherence to physical constraints

D. 3D Model Stylization and Texture Generation

Existing 3D model stylization works, like StyleCity [14] and TEXTure [15], focus on "reskinning" a fixed geometry. In contrast, our task is not style "transfer" but style "generation" from scratch. More importantly, our goal is not merely to match an arbitrary artistic style but to inherit and reproduce the intrinsic architectural features extracted from the original CIM model, making it a deeper, more constrained generation task.

III. METHODOLOGY

This section details the proposed knowledge-guided, feature-continuous regeneration framework. We first provide an overview of the entire pipeline, followed by a deep dive into the core technologies and implementation details of each stage

A. System Overview: The Deconstruct-Generate-Reconstruct-Render Pipeline

As shown in Figure 1, the core of our methodology is an automated four-stage process: "Deconstruct-Generate-Reconstruct-Render." These four tightly integrated stages communicate and are constrained by a central "visual

knowledge fingerprint," forming a complete generation chain from abstract knowledge to a concrete 3D model. This fingerprint acts as a structured, machine-readable representation of the original building's essence, serving as the semantic bridge that ensures coherence and controllability throughout the creative process.

1) **Deconstruct:** This stage serves as the knowledge source. It takes an original CIM building model as input and, through a series of analysis algorithms, "digests" it into a structured, multi-dimensional "visual knowledge fingerprint."

2) **Generate:** This stage is responsible for creating a new 2D form. It retrieves style information and plot boundary constraints from the knowledge fingerprint and uses our specially trained Arch-Layout LoRA model to generate a new 2D building footprint that is architecturally sound and stylistically resonant with the original building.

3) **Reconstruct:** This stage elevates the 2D form into a 3D structure. Based on the 2D footprint from the second stage, it reads physical constraints and the semantic composition vector from the knowledge fingerprint. A procedural engine, dynamically parameterized by this knowledge, constructs a 3D massing model that satisfies all constraints and simultaneously generates a semantic ID map for its surfaces.

4) **Render/Paint:** This stage endows the 3D "skeleton" with vivid "flesh and blood." It uses the geometry and semantic ID map from the third stage, combined with the

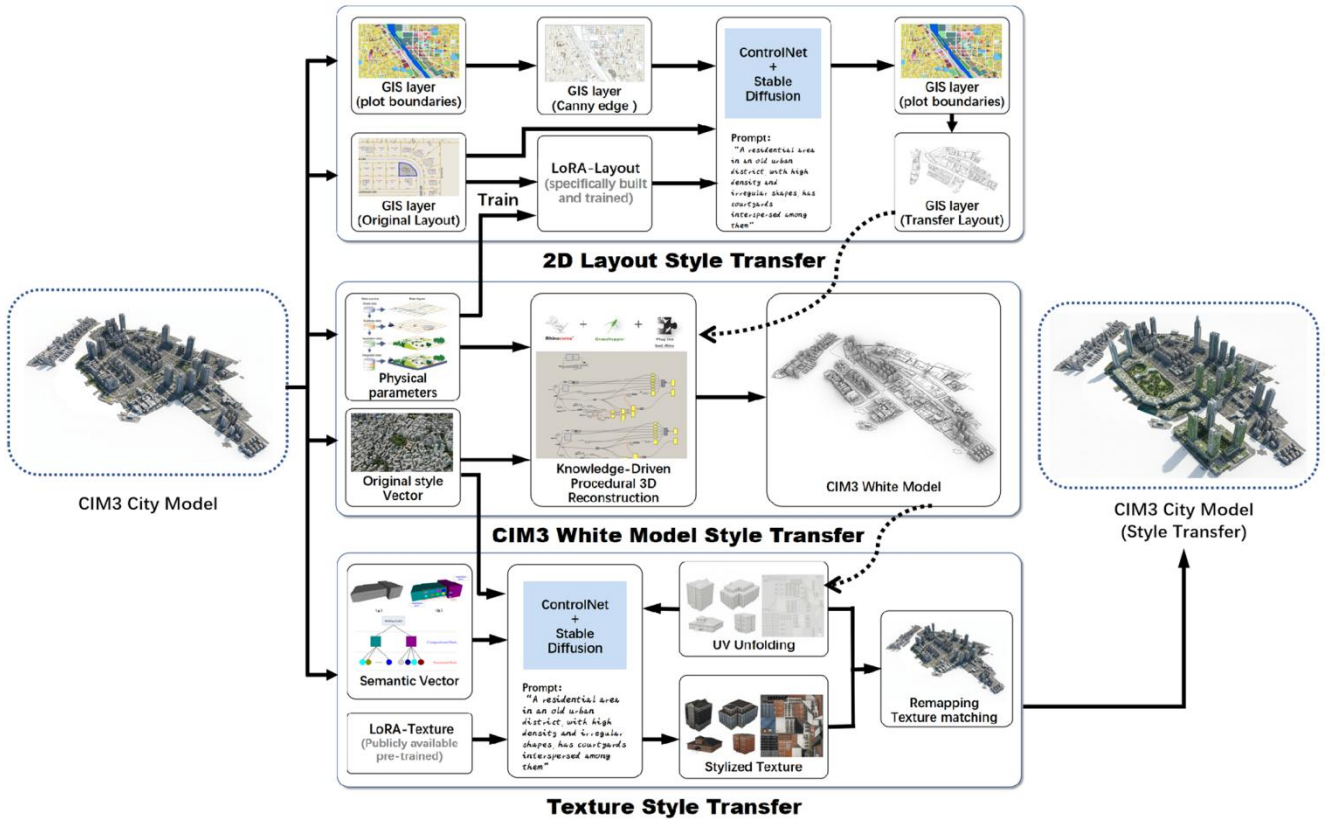


Fig. 1. Overall flowchart

semantic composition and style vectors from the knowledge fingerprint, to generate high-fidelity textures for the model's surface via a multi-modally guided diffusion model.

Throughout this pipeline, the knowledge fingerprint acts as both a data hub and a constraint center. It transforms unstructured raw model information into structured, computable instructions, ensuring that outputs of the subsequent three creative stages remain logically coherent and consistent.

B. Stage One: CIM Deconstruction and Visual Knowledge Fingerprint Extraction

The initial stage of our pipeline is dedicated to the deconstruction of the source CIM model to extract a "visual knowledge fingerprint." This fingerprint serves as a structured, multi-faceted representation of the building's essential characteristics. Its components were deliberately chosen to form an orthogonal basis for architectural identity, encompassing programmatic constraints, material composition, and aesthetic character. The fingerprint comprises three distinct parts. First, Physical Parameter Extraction establishes immutable hard constraints for the design. Parameters, such as Plot Area (A_{plot}), Total Floor Area (A_{floor}), Floor Area Ratio (FAR), and overall Building Height (H), are calculated directly from the geometric and geographic data embedded in the CIM model. Second, a Semantic Composition Vector (v_{sem}) quantifies the proportional distribution of functional and material components across the building's facade. To achieve this, the original model is rendered from multiple viewpoints and processed by a pre-trained semantic segmentation model (e.g., SegFormer [16]). By calculating the normalized pixel count for each semantic category—such as wall, window, door, and roof—we construct a vector that precisely represents the area proportions of these key architectural elements. Finally, a Style Vector (v_{style}) is derived to capture the building's macro-level artistic style. Material textures from the CIM are encoded using the image encoder of a pre-trained vision-language model, CLIP [2]. The resulting high-dimensional feature vector effectively encapsulates core stylistic attributes, including material, texture, and color, by leveraging CLIP's learned feature space, which aligns visual concepts with natural language. To handle buildings with multiple materials, all associated textures are first composited into a single texture atlas. This atlas is then processed by CLIP's image encoder to generate a holistic vector that captures the building's overall material and aesthetic character.

Crucially, this fingerprint is not derived solely from 3D geometry. It represents a fusion of multi-source urban knowledge. For instance, the physical parameters such as Floor Area Ratio (FAR) are cross-validated against official planning documents, including local control plans and construction plans. This integration ensures the extracted knowledge is not merely descriptive but also prescriptive, conforming to real-world regulatory constraints.

C. Stage Two: AI-Driven 2D Architectural Layout Generation

The second stage introduces the first generative process, producing a novel, geometrically valid, and stylistically

continuous 2D building footprint. This is achieved using a purpose-built module we term Arch-Layout LoRA. Recognizing that general-purpose diffusion models often produce organic, non-orthogonal shapes unsuitable for practical architecture, we injected critical domain knowledge using Low-Rank Adaptation (LoRA), a parameter-efficient fine-tuning technique. The success of our Arch-Layout LoRA hinges on a high-quality, domain-specific dataset. We constructed this dataset by extracting building footprints directly from the Shenzhen CIM3-level city model, ensuring alignment with the real-world architectural characteristics and urban context targeted by our pipeline.

To facilitate the creation of rich text descriptions for this dataset, we designed a novel Semi-Automated Annotation Pipeline. This pipeline provides a Multimodal Large Language Model (e.g., GPT-4V) with a combination of inputs: (1) the binary image of the building footprint extracted from the Shenzhen CIM3 model, and (2) textual metadata derived from corresponding planning documents, specifying attributes like land use, FAR limits, and height restrictions. Through a carefully engineered prompt, the MLLM is guided to generate a rich, natural language description. The prompt encourages the model to describe key architectural features evident in the footprint and implied by the planning data, such as building morphology (e.g., 'L-shaped', 'courtyard-style'), layout patterns, typical roof forms, and functional zoning. The Arch-Layout LoRA was subsequently trained on a pre-trained Stable Diffusion v1.5 model using approximately 50,000 of these image-text pairs. To validate the outputs, we designed custom evaluation metrics:

A) **Geometric Validity**, measuring the percentage of valid polygons;

B) **Constraint Compliance Rate**, calculated as the IoU with the input plot mask;

C) **Architectural Plausibility**, a heuristic score penalizing excessive vertices or small acute angles. In the final conditional generation process, Arch-Layout LoRA is synergistically paired with ControlNet [10], where the target plot's boundary serves as a strict spatial mask and keywords decoded from the knowledge fingerprint. This decoding is achieved via a predefined mapping dictionary that translates clusters in the style vector space to descriptive tags (e.g., 'symmetrical', 'glass facade'). A text prompt, constructed from a general architectural description and keywords decoded from the knowledge fingerprint, guides the diffusion model's denoising process. This dual guidance ensures the generated shape is both geometrically confined and architecturally characteristic, with the final raster output being vectorized into clean polygon data.

D. Stage Three: Knowledge-Driven Procedural 3D Reconstruction

In the third stage, the 2D footprint generated previously is extruded into a 3D massing model that rigorously conforms to the original building's core features. This is achieved via a deterministic procedural reconstruction engine dynamically driven by the knowledge fingerprint. This hybrid methodology leverages the complementary strengths of different paradigms: AI for high-level, fuzzy pattern recognition (style) and

procedural systems for deterministic, rule-based precision (geometry). By delegating high-level decisions to the AI component and execution to the procedural engine, our system avoids the geometric imprecision of purely generative 3D models while overcoming the creative rigidity of purely procedural approaches. The engine's operation is guided by the fingerprint in several ways. First, a Style-Driven Rule Selection mechanism uses the style vector (v_{style}) to choose an appropriate procedural rule library. Pre-defined libraries, each corresponding to a macro architectural style (e.g., modernist.cga), are compared with the input (v_{style}) using cosine similarity, and the best-matching library is selected. Concurrently, Semantic-Driven Parameterization uses the semantic composition vector (v_{sem}) to dynamically set the parameters within these rules. For instance, if the 'window' proportion in v_{sem} is 0.25, a facade-splitting rule's ratio parameters are automatically adjusted to produce a facade with approximately 25% window area. To ensure Constraint Adherence, the engine incorporates an iterative solver that first generates an initial model, then checks its FAR against the target, and iteratively adjusts parameters like building height until the error falls within a predefined tolerance. Critically, as the engine constructs the geometry, it also generates a Semantic ID Map by assigning a unique label to each face, which is then baked into a texture map to serve as a structured input in the final stylization stage.

E. Stage Four: Knowledge-Guided Appearance Stylization and Rendering

The final stage imbues the reconstructed 3D model with a realistic appearance that maintains feature continuity with the source building. This is accomplished through a sophisticated image generation process based on a pre-trained diffusion model, heavily constrained by a multi-modal guidance mechanism to ensure both fidelity and quality. The primary constraint is Structural Guidance, which guarantees perfect alignment between the generated texture and the underlying geometry. The semantic ID map created in the previous stage is fed into a ControlNet-Seg model, compelling the diffusion model to generate semantically appropriate content exclusively within designated regions (e.g., generating glass textures only in 'window' areas), thereby preventing common artifacts like color bleeding. This is complemented by a softer, content-level Semantic Guidance, where the semantic composition vector (v_{sem}) is translated into a descriptive text prompt (e.g., "a photorealistic facade of a brick and glass building"). This guides the model to generate the correct material types in their intended proportions. Finally, Style Guidance determines the aesthetic "feel" of the rendered texture. We employ a hybrid mechanism that combines internal and external guidance. Internal guidance uses the original style vector (v_{style}) to steer the output toward the source model's aesthetic tone. In contrast, external guidance allows users to specify community-trained style LoRAs (e.g., a "cyberpunk" style) to inject a completely new artistic vision. This modular design enables users to "plug and play" various styles, dramatically enhancing the system's flexibility and applicability across diverse creative domains.

IV. EXPERIMENTS AND RESULTS

To validate the effectiveness of our proposed knowledge-guided regeneration framework, we conducted a series of qualitative and quantitative experiments.

A. Experimental Setup

Experiments used a CIM3-level 3D city model dataset of Shenzhen as the source data. The training dataset for Arch-Layout LoRA was constructed from approximately 50,000 building footprints derived from this primary Shenzhen CIM3-level 3D city model dataset. The pipeline was implemented in Python, PyTorch, and the Hugging Face Diffusers library, and ran on an NVIDIA A100 GPU. Base models included Stable Diffusion v1.5, SegFormer-B5 [16], and OpenAI's CLIP ViT-L/14 [2]. Evaluation metrics included Geometric Validity (%) and Constraint Compliance Rate (IoU) for the layout generator, and Fréchet Inception Distance (FID), Semantic Consistency (cosine similarity of v_{sem}), Style Consistency (cosine similarity of CLIP embeddings), and Physical Constraint Adherence (relative error of FAR) for the end-to-end generation.

B. Performance Evaluation of Arch-Layout LoRA

Arch-Layout LoRA is key to ensuring the plausibility of the generated geometric forms. Its performance was compared against a baseline Stable Diffusion v1.5 model. Qualitative comparison (as shown in Figure 2) shows that the baseline model often generated organic, irregular shapes, whereas the fine-tuned model produced footprints with a much stronger preference for straight lines and right angles, aligning with real-world building characteristics. To quantify this, 1,000 samples were generated by both models. The results, shown in Table II, provide strong evidence for the necessity and effectiveness of the dedicated AI toolchain. Arch-Layout LoRA achieved a decisive advantage in geometric validity and demonstrated higher precision in adhering to plot boundary constraints.

TABLE II. PERFORMANCE COMPARISON OF 2D LAYOUT GENERATION MODELS

<i>Model</i>	<i>Geometric Validity (%)</i>	<i>Constraint Compliance (IoU)</i>
<i>Baseline SD v1.5</i>	44.7	0.81
<i>Our Arch-Layout LoRA</i>	98.2	0.95



Fig. 2. Qualitative Comparison between Baseline SD v1.5 and Our Arch-Layout LoRA

C. End-to-End Generation Results

Complete generation cases visually demonstrate the end-to-end capabilities of the pipeline. It can be clearly seen from the example in Figure 3, the examples clearly show that the method successfully achieves its core objective: precisely inheriting the physical, semantic, and stylistic essence of the original building while completely altering its geometric entity to ensure privacy and security.

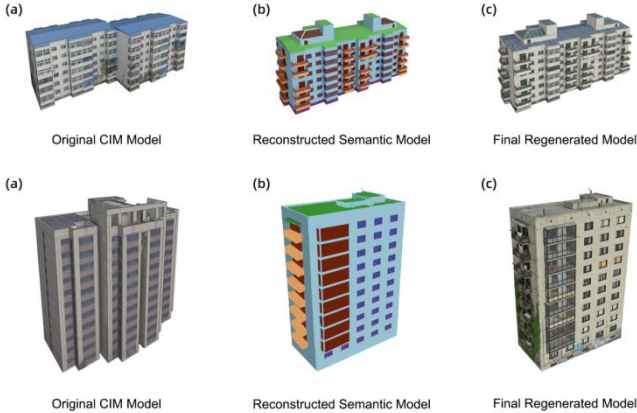


Fig. 3. End-to-End Generation Results and Feature Continuance Demonstration

D. Ablation Study

To verify the independent contributions of each component within the knowledge fingerprint, a rigorous ablation study was designed. The full model was used as a baseline, and semantic guidance, style guidance, and physical constraints were systematically removed. The results, averaged over 20 buildings, are presented in Table III. The findings from this study do more than validate components; they reveal the internal mechanics of the model. For example, the substantial drop in semantic consistency when semantic guidance is removed proves that text prompts alone are insufficient for controlling detailed material distribution in complex scenes—a key finding for the broader controllable generation community.

Similarly, the skyrocketing FAR error without physical constraints highlights the continued relevance of structured solvers in an era of deep learning, showing that hybrid models are currently superior for tasks requiring hard constraint satisfaction.

1) **Removing Semantic Guidance:** This led to a substantial 28.6% drop in semantic consistency, manifesting as incorrect material proportions. This confirms the critical role of the semantic composition vector in guiding both procedural reconstruction and texture generation.

2) **Removing Style Guidance:** Style consistency dropped by 18.2%, weakening the aesthetic connection between the new and original buildings. This validates the indispensable role of the CLIP style vector in selecting the appropriate procedural rule library and guiding the texturing process.

3) **Removing Physical Constraints:** The relative error for FAR skyrocketed by over 1200%, indicating that the generated buildings were completely out of control in terms of volume. This highlights the core value of the iterative solver in ensuring compliance with real-world physical constraints.

This study provides compelling evidence that the proposed "knowledge fingerprint" is a well-designed system with orthogonal and complementary components, each making a critical contribution to the final result.

TABLE III. ABLATION STUDY OF THE PROPOSED FRAMEWORK

<i>Model Configuration</i>	<i>FID</i> ↓	<i>Semantic Consistency</i> ↑	<i>Style Consistency</i> ↑	<i>FAR Rel. Error (%)</i> ↓
<i>Full Model</i>	25.4	0.91	0.88	1.2
<i>w/o Semantic Guidance</i>	31.2	0.65	0.87	1.5
<i>w/o Style Guidance</i>	29.8	0.89	0.72	1.3
<i>w/o Physical Constraints</i>	26.1	0.9	0.88	15.8

E. Application Case: Stylistic Flexibility

To showcase the pipeline's potential as a general-purpose content generation tool, an application case study was conducted. As shown in Figure 4, a generated classical residential building was used as a base, and different external style LoRA modules were swapped in during the rendering stage. The results demonstrate the high degree of modularity and extensibility of the pipeline. By simply "plugging and playing" different style LoRAs, users can easily impart a vast array of artistic styles to the same base model, showcasing its potential in creative content generation for industries like gaming, film, and the metaverse.



Fig. 4. Application Case for Stylistic Flexibility using Pluggable LoRA Modules

V. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper addresses the dual challenges of privacy security and data asset diversification in CIM data sharing by proposing an innovative solution: "Knowledge-Guided Feature-Continuous Regeneration." We constructed and validated a four-stage hybrid AI generation pipeline — "Deconstruct-Generate-Reconstruct-Render" — whose core lies in using a structured "visual knowledge fingerprint" to guide the generation of new 3D building models. This approach enables controllable inheritance of the original building's core features while completely reshaping its geometry and appearance. Key contributions include: a complete theoretical framework for knowledge inheritance, a dedicated AI toolchain for architectural layout generation (Arch-Layout LoRA), a hybrid intelligent reconstruction paradigm that combines AI guidance with procedural precision, and a multi-modal guided texture generation mechanism. In summary, the proposed method not only provides a foundational solution to CIM data desensitization that transcends traditional "subtractive" approaches, but also advances the development of controllable, high-quality generative AI for 3D content creation.

B. Limitations and Future Work

Despite encouraging results, the method has limitations, including dependence on the procedural rule library for morphological richness and high computational overhead. These limitations open several exciting avenues for future research. As conditional 3D generation technology matures, a promising direction is to replace the procedural reconstruction and rendering stages with a powerful, end-to-end conditional 3D foundation model. The "visual knowledge fingerprint" and 2D footprint could serve as a unified multi-modal condition to generate models with even greater morphological diversity. The current pipeline could also be extended from single buildings to entire city blocks, using Scene Graphs as a higher-

level knowledge representation to guide the collaborative construction of coherent urban scenes. Finally, this pipeline could be developed into an interactive creation tool, positioning AI not as an automated black box but as an intelligent co-pilot for architects. In such a system, the knowledge fingerprint becomes a dynamic, user-controllable interface for exploring the design space, ushering in a new era of human-computer collaborative design.

REFERENCES

- [1] T. H. Kolbe, "Representing and exchanging 3D city models with CityGML," in 3D Geo-Information Sciences. Springer, 2009, pp. 15–31.
- [2] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in Proc. Int. Conf. Mach. Learn. (ICML), 2021, pp. 8748–8763.
- [3] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in Proc. Adv. Neural Inf. Process. Syst. (NIPS), 2012, pp. 1097–1105.
- [4] F. Biljecki, H. Ledoux, and J. Stoter, "Generation of multi-LOD 3D city models in CityGML with the procedural modelling engine Random3Dcity," ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci., vol. IV-4/W1, pp. 51–59, 2016.
- [5] P. Müller, P. Wonka, S. Haegler, A. Ulmer, and L. Van Gool, "Procedural modeling of buildings," ACM Trans. Graph., vol. 25, no. 3, pp. 614–623, Jul. 2006.
- [6] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in Proc. IEEE Conf. Comput. Vis. Pattern Recog. (CVPR), 2017, pp. 1125–1134.
- [7] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), 2017, pp. 2223–2232.
- [8] N. Nauata, S. Hosseini, K.-H. Chang, H. Chu, C.-Y. Cheng, and Y. Furukawa, "House-GAN++: Generative adversarial layout refinement networks," in Proc. IEEE Conf. Comput. Vis. Pattern Recog. (CVPR), 2021, pp. 14008–14017.
- [9] K.-H. Chang, R. Zhang, L. Yi, S. Bai, and D. G. Aliaga, "Building-GAN: Graph-conditioned architectural volumetric design generation," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), 2021, pp. 1544–1553.
- [10] L. Zhang, A. Rao, and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), 2023, pp. 3836–3847.
- [11] H. Jun and A. Nichol, "Shap-E: Generating conditional 3D implicit functions," arXiv preprint arXiv:2305.02463, 2023.
- [12] B. Poole, A. Jain, J. T. Barron, and B. Mildenhall, "DreamFusion: Text-to-3D using 2D diffusion," arXiv preprint arXiv:2209.14988, 2022.
- [13] H. Wang, X. Du, J. Li, R. A. Yeh, and G. Shakhnarovich, "Score Jacobian chaining: Lifting pretrained 2D diffusion models for 3D generation," in Proc. IEEE Conf. Comput. Vis. Pattern Recog. (CVPR), 2023, pp. 12595–12605.
- [14] Y. Chen, S. Lai, Z. Liu, L. Bao, Q. Li, Z. Liu, J. Yuan, and Y.-J. Liu, "StyleCity: Large-scale 3D urban scenes stylization with vision-and-text reference via progressive optimization," arXiv preprint arXiv:2404.10681, 2024.
- [15] E. Richardson, G. Metzer, Y. Alaluf, R. Giryes, and D. Cohen-Or, "TEXTure: Text-guided texturing of 3D shapes," ACM Trans. Graph., vol. 42, no. 4, art. 53, 2023.
- [16] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2021, pp. 12077–12090.